IWATANI CORPORATION · CONFIDENTIAL WORKING DRAFT

Value Chain Bibliography

Every website, company page, brochure and filing used to build the 12-stage bellows value chain, organised by stage

Client

Iwatani Corporation — Metals Department → Stainless Steel Division

Subject

Bellows tube (precision-slit material) market entry, leveraging the Jindal Stainless relationship

Status

Working document — updated at monthly internal meetings

Prepared

July 2026

Evidence standard. Every factual claim in this document carries an inline source link and a verification grade: **[P]** primary (company, association or government publication), **[C]** credible press, **[S]** commercial market-research vendor (indicative only), **[U]** unverified directory listing. Commercial estimates for the bellows market disagree with one another by up to a factor of 19; figures marked **[S]** must not be quoted externally without stating the range. Companion to the topic-organised source register in file 06.

17 — Value Chain Bibliography: Every Source, by Stage

Every website, company page, brochure, and corporate filing used to build the 12-stage value chain map (`14-value-chain-map.md`) and the technical/competitive material behind it (`01`, `04`, `07`, `08`), organised **by the stage of the chain each source supports** rather than alphabetically. Cross-check any player name on charts 27–29 against the stage below.

This is a companion to `06-source-register.md`, which lists the same sources organised by topic (technology, market sizing, India policy, etc.) rather than by chain position. Use this file when you want to know "where did stage X come from"; use file 06 when you want to know "where did claim Y come from."

Grades: [P] company/primary · **[C]** credible press · **[S]** commercial market research (indicative) · **[U]** unverified directory/listing.

Stage 1 — Raw materials (ore, ferrochrome, nickel)

Source	URL	Grade	What it supports
Jindal Stainless — Stainless Steel Product Brochure	https://www.jindalstainless.com/product-brochure/	**[P]**	Sukinda chrome ore mines (Odisha); Ferro Chrome plant 250,000 tpa at Jajpur; ferro alloy furnace specs (SMS DeMAG, Kopern Maco)

Global raw-material players (Glencore, Vale, Nornickel, Eramet, Tsingshan, South32, Yildirim) are named on chart 27 as standard industry references for chrome/nickel/molybdenum supply and are not individually sourced — they are the recognised majors in ferroalloy and battery-metal mining, included for completeness of the map rather than for a specific claim.

Stage 2 — Melt & cast (EAF → AOD → VOD → slab)

Source	URL	Grade	What it supports
Jindal Stainless — Product Brochure	https://www.jindalstainless.com/product-brochure/	**[P]**	Hisar melt shop (2 EAF/induction + 2 AOD + 1 VOD, 800,000 tpa); Jajpur melt shop design (SMS Siemag), liquid ferrochrome process, 1.1 Mtpa (→2.9 Mtpa planned)
Jindal Stainless — Annual Report FY2023-24	https://www.jindalstainless.com/annualreport/2023-2024/corporate-overview/manufacturing-strengths	**[P]**	16 facilities across India/Spain/Indonesia; Hisar 0.8 Mtpa, Jajpur 2.2 Mtpa
Business Standard — JSHL brownfield expansion	https://www.business-standard.com/article/news-cm/jindal-stainless-hisar-commissions-26-000-tpa-capacity-precision-strip-mill-121102100487_1.html	**[C]**	Independent corroboration of Hisar melt capacity (800,000 tpa)
Hindu BusinessLine — Jindal ₹2,600 cr capex	https://www.thehindubusinessline.com/companies/jindal-stainless-eyes-2600-cr-capex-to-double-melting-capacity/article66226105.ece	**[C]**	Melt capacity expansion to 2.9 Mtpa; management attribution (Abhyuday Jindal)
ATI — AM 350 / AM 350-MIL	https://www.atimaterials.com/Products/am-350	**[P]**	AM350 melt chemistry (Cr 16–17%, Ni 4–5%, Mo 2.5–3.25%, N 0.07–0.13%) — the grade **no Indian mill melts**
Lamineries Matthey — AM350 data sheet	https://www.matthey.ch/fileadmin/user_upload/downloads/fichetechnique/EN/Inox-AM350_v25E.pdf	**[P]**	AM350 as a "maraging" semi-austenitic PH grade requiring specific melt/heat-treat control; semiconductor named as an end application
TOKKIN — TOKKIN 350	https://www.tokkin.com/search/stainless-steels/precipitation-hardening/tokkin350/	**[P]**	Benchmark for a mill that explicitly melts and markets a bellows-grade PH alloy — the standard Jindal is not currently meeting

Stage 3 — Hot roll

Source	URL	Grade	What it supports
Jindal Stainless — Product Brochure	https://www.jindalstainless.com/product-brochure/	**[P]**	Hisar Hot Steckel Mill (720,000 tpa) + Tandem Strip Mill (300,000 tpa); Jajpur 1.0 Mtpa Tandem Mill (Siemens VAI), Five Stein reheating furnace

Global integrated mills (Outokumpu, Acerinox, Aperam, POSCO, Nippon Steel Stainless, NAS, Tsingshan, ATI, Carpenter) named on chart 27 are standard-reference global stainless producers at this stage; not individually sourced for a specific claim beyond their well-known market position.

Stage 4 — Cold roll to thin gauge ★ core to the whole thesis

Source	URL	Grade	What it supports
Jindal Stainless — Product Brochure	https://www.jindalstainless.com/product-brochure/	**[P]**	Cold Rolling Complex (375,000 tpa, four 20-Hi Sendzimir mills); Speciality Products Division (84,000 tpa precision CR strip, razor blade strip **to 0.076 mm**); grade tables — **AM350/631 absent**
Jindal Stainless — JSHL brownfield expansion press release	https://www.jindalstainless.com/press-releases/jindal-stainless-hisar-limited-commissions-phase-i-of-brownfield-expansion-at-specialty-products-division/	**[P]**	Precision strip capacity path 22,000→48,000→60,000 tpa; blade steel **to 0.07 mm**; ~650 mm width; ~₹450/250/200 cr capex
Prime Stainless — JSL products	https://primestainless.com/products/	**[C]**	Precision strips rolled to 0.05 mm; razor blade strip to 0.076 mm at JSL Hisar SPD
IUP Jindal Metals & Alloys — brochure	https://jindalmetal.com/images/iup-brochure.pdf	**[P]**	**0.03–1.5 mm** thickness range; 20-Hi mill with AGC; separate Jindal SAW group company
IUP Jindal — product page	https://jindalmetal.com/product-range/stainless-steel-manufacturers-in-india-2/cold-rolled-precision-stainless-steel-strips	**[P]**	22,000 tpa capacity; grade list (200/300/400 series); tolerance/finish specs
IUP Jindal — infrastructure & facilities (IndiaMART)	https://www.indiamart.com/iup-jindal-metals/infrastructure-and-facilities.html	**[U]**	Three Sendzimir mills, I2S AGC, Landis roll grinding, two bright annealing lines, mill capability table
Quality Foils (India) — about us	https://www.qualitygroup.in/qualityfoils/about-us/	**[P]**	Cold rolling since 1982, Hisar; width 20–710 mm, thickness 0.10–4.00 mm
Quality Foils — products	https://www.qualitygroup.in/qualityfoils/products/	**[P]**	Published thickness tolerance table (**±0.020 mm at 0.10 mm gauge**) — the critical tolerance-gap data point
Kompass — Quality Foils listing	https://www.kompass.com/z/ww/c/quality-foils-india-private-limited/in756895/	**[C]**	Confirms "Metallic Bellows" and "Bellows for precision measuring instruments" in Quality Foils' product range
Alleima — precision strip steel brochure	https://www.alleima.com/siteassets/documents/strip/precision_strip_steel_brochure_fin.pdf	**[P]**	Global benchmark: strip to **0.015 mm at ±0.001 mm tolerance**; PH steels and nickel alloys in spring-strip programme; "thermostat expansion bellows" named
Alleima — strip steel product page	https://www.alleima.com/en/products/strip-steel/	**[P]**	Full precision strip portfolio; distributor-expansion statement
Alleima — distributors page	https://www.alleima.com/en/products/strip-steel/distributors/	**[P]**	No current Indian distributor listed — a strategic gap/opening noted in file 08
Alleima — list of Alleima entities	https://www.alleima.com/en/about-this-site/data-privacy-portal/list-of-alleima-entities/	**[P]**	Confirms Alleima India Pvt Ltd, Pune address
TheCompanyCheck — Alleima India financials	https://www.thecompanycheck.com/company/alleima-india-private-limited/U29308PN2019PTC182454	**[U]**	Formerly Sandvik Materials Technology India; ~125 employees; incorporated 2019
LinkedIn — Shailesh Sardesai (Alleima India)	https://linkedin.com/in/shailesh-sardesai-6635514	**[U]**	VP & Regional Sales Director, Strip Division India; confirms India segment focus (compressor valves, razor blades, springs — no bellows listed)
Ulbrich — AM 350 stainless steel	https://www.ulbrich.com/alloys/am-350-stainless-steel-uns-s35000/	**[P]**	AM350 in strip/coil/foil to AMS 5548/ ASTM A693; **"Bellows" named as the first application**
United Performance Metals — AM 350	https://www.upmet.com/products/stainless-steel/am-350r	**[P]**	AM350 precision reroll strip **0.0008"–0.015"**; bellows named under aerospace applications

Source	URL	Grade	What it supports
YES Stainless — precision strip	https://yesstainless.com/products/stainless-steel-precision-strip/	**[U]**	Burr control methodology; 0.02–1.5 mm range
GoldSupplier — precision slit strip listings	https://www.goldsupplier.com/provide/p173578899.html · https://www.goldsupplier.com/provide/p173487071.html	**[U]**	Typical precision slitting tolerance bands quoted by a commercial supplier
Sachiya Steel International	https://www.sachiyasteelintl.com/stainless-steel-precision-strip.html	**[U]**	Indian stockist-processor capability, 0.02–4.0 mm
Grotal — Precision Stainless Steel Strip Manufacturers, Delhi	https://www.grotal.com/Delhi/Precision-Stainless-Steel-Strip-Manufacturers-C44/	**[U]**	Directory confirming Hisar Metal Industries, Singhal Strips as regional strip mills

Stage 5 — Temper, level, finish

No dedicated new sources — this stage draws on the same mill pages already listed under Stage 4 (Jindal SPD's strip grinding/skin pass/tension leveller; IUP Jindal's bright annealing lines and Brodeur tension leveller). Iwatani's own spring/gasket temper products are sourced under Stage 6.

Stage 6 — Precision slitting ★ the proposed entry point

Source	URL	Grade	What it supports
Iwatani — Stainless Steel (Materials/Metals)	https://www.iwatani.co.jp/eng/business/material/metals/products/stainless/	**[P]**	**Precision Slitting Sites: Suzhou Iwatani Metal Products & Zhongshan Iwatani (China)**; ultrathin foil to **8 µm**; spring/gasket/hydrogen-resistant grade lines
Iwatani — FY2026 Investors' Guide	https://www.iwatani.co.jp/eng/ir/pdf/about_iwatani.pdf	**[P]**	Thailand precision pressing plant (2025 addition); integrated procurement-to-processing model precedent
Hempel Special Metals — precision slit strip	https://www.hempel-metals.com/en/products/precision-slit-strip	**[P]**	Service centre explicitly serving **metallic bellows and expansion joint manufacturers**, from 0.05 mm
IUP Jindal — brochure	https://jindalmetal.com/images/iup-brochure.pdf	**[P]**	Three Brodeur slitting lines, shimless tooling, edge-rounding machine; application list includes **"Flexi metal tubes / bellows"**
Quality Foils — Kompass listing	https://www.kompass.com/z/ww/c/quality-foils-india-private-limited/in756895/	**[C]**	Slit edge from 5.0 mm; **"Metallic Bellows"** named in product range
Lamineries Matthey — AM350 data sheet	https://www.matthey.ch/fileadmin/user_upload/downloads/fichetechnique/EN/Inox-AM350_v25E.pdf	**[P]**	AM350 strip width tolerance to **±0.1 mm on request** — a named global precision slitter for this specific grade
SS-Strips.com — SS316L for vacuum bellows listing	https://www.ss-strips.com/sale-36582674-flexible-ss316l-cold-rolled-precision-stainless-steel-strip-for-vacuum-bellows-0-11-79mm.html	**[U]**	Chinese re-roller openly marketing "strip for vacuum bellows" at 0.11 mm; burr spec; 3-tonne MOQ
Aesteiron — AM350/UNS S35000	https://www.aesteiron.com/alloym350.html	**[P]**	Mumbai-based importer/stockist quoting AM350 strip/plate prices — confirms an existing Indian trader channel for the grade

Stage 7A — Formed / convoluted bellows manufacture

Source	URL	Grade	What it supports
Triad Bellows — how metal bellows are made	https://triadbellows.com/how-metal-bellows-are-made	**[P]**	Full process: shear → roll to tube → longitudinal seam weld → hydroform/ mechanical form; multi-ply telescoping
Comflex — manufacturing process of metal expansion joints	https://www.metalhosemachine.com/purchase-guide/manufacturing-process-of-metal-expansion-joints/	**[C]**	TIG/plasma longitudinal welding; hydraulic vs mechanical corrugation forming
Pipeline Dubai — U-shaped multilayer bellows QC	https://www.pipelinedubai.com/manufacturing-quality-control-of-u-shaped-multilayer-metal-bellows-expansion-joint.html	**[C]**	Two-stage hydroforming mechanics; pressure-control failure modes
Intran — advantages of hydroformed bellows	https://www.intran.mx/advantages-of-hydroformed-bellows/	**[C]**	Molded vs welded taxonomy
ScienceDirect — single-step tube hydroforming	https://www.sciencedirect.com/science/article/abs/pii/S0924013607001562	**[P]**	Conventional four-step forming route vs single-step hydroforming
Witzenmann India — brochure	https://media.witzenmann.com/mediapool/documents/brochures/witzenmann-india.pdf	**[P]**	HYDRA corrugated bellows process (hydraulic forming of thin-walled tubes); single-ply vacuum vs multi-wall high-pressure use
Flexpert Bellows	https://flexpertbellows.com/	**[P]**	Belgaum manufacturer; ISO 9001; EJMA; blue-chip client list
Flexpert — metal expansion joints	https://www.flexpertbellows.com/metal-expansion-joints/	**[P]**	Multi-ply laminated construction, 2–20 plies
MB Metallic Bellows Pvt Ltd	https://mbmetallicbellows.com/	**[P]**	Large-bore expansion joints, 100 NB to 8 m diameter, ASME U-stamp
Metallic Bellows (India) Pvt Ltd	https://metallicbellows.com/	**[P]**	Company history (incorporated 1980), founder background
Triveni Engineers	https://www.triveniengineers.com/expansion-bellows.html	**[U]**	Ahmedabad regional fabricator

Stage 7B — Edge-welded bellows manufacture ★ semiconductor route

Source	URL	Grade
MW Components — edge welding technology	https://www.mwcomponents.com/process/edge-welding-technology	**[P]**
Technetics Group — metal bellows	https://technetics.com/technetics-products/metal-bellows/	**[P]**
Technetics — BELFAB edge welded bellows	https://technetics.com/products/belfab-edge-welded-metal-bellows/	**[P]**
KSM USA — edge welded metal bellows	https://www.ksmusa.com/edge-welded-metal-bellows	**[P]**
KSM — SEMICON West 2026 exhibitor profile	https://expo.semi.org/west2026/Public/eBooth.aspx?BoothID=658784&FromPage=Exhibitors.aspx&IndexInList=310&ListByBooth=true&Nav=False&ParentBoothID=	**[P]**
Metal Flex Welded Bellows	https://www.metalflexbellows.com/	**[P]**
Irie Koken — semiconductor field	https://www.ikc.co.jp/en/field/semiconductor.html	**[P]**
Irie Koken — about	https://www.ikc.co.jp/en/about.html	**[P]**
MIRAPRO — company profile	https://www.mirapro.co.jp/en/company/profile/	**[P]**
Eagle Industry (EKK) — welded metal bellows	https://www.ekkeagle.com/en/product/welded-metal-bellows	**[P]**
Valqua — bellows	https://www.valqua.com/product_classification/bellows/	**[P]**

Source	URL	Grade
Well Tech Metal Bellows — IndiaMART company page	https://www.indiamart.com/welltechmetal-bellowsi/	**[U]**
Well Tech — mechanical seal bellows listing	https://www.indiamart.com/proddetail/edge-welded-metal-bellows-for-mechanical-seal-assemblies-2859074851312.html	**[U]**
Well Tech — UHV bellows listing	https://www.indiamart.com/proddetail/edge-welded-metal-bellows-for-vacuum-chambers-uhv-systems-2859074772533.html	**[U]**
LinkedIn — Well Tech Metal Bellows	https://linkedin.com/in/well-tech-metal-bellows-i-pvt-ltd-well-tech-32818625b	**[U]**
Fluidyne Engineers — homepage	https://fluidyneengineers.com/	**[P]**
Fluidyne — diaphragms, capsules & edge welded bellows	https://fluidyneengineers.com/Diaphragm_Seals_Capsules.php	**[P]**
Fluidyne Engineers India — our products	https://www.fluidyneengineersindia.com/our-products.html	**[P]**
Fluidyne — diaphragm page	https://www.fluidyneengineersindia.com/diaphragm.html	**[P]**
Bhastrik Mechanical Labs — TamilCrew profile	https://tamilcrew.com/engineering/industrial/bellows/	**[U]**
	https://www.tradeindia.com/products/edge-welded-bellows-614236.html	**[U]**

Source	URL	Grade	W
Bhastrik — TradeIndia product listing			

Stage 8 — Assembly & hardware

Source	URL	Grade	What it supports
Hind High Vacuum (HHV)	https://hhv.in/	**[P]**	61-year vacuum technology group; thin-film systems, furnaces, optics
Fourvac Technologies	https://fourvac.com/	**[P]**	UHV chambers, XYZ manipulators, drives and motions; IIT Kanpur six-chamber UHV system
Advanced Process Technology (APT)	https://aptglobal.com/vacuum-chambers/	**[P]**	Custom HV/UHV chambers; leak tested to 1×10^{-12} Torr·l/s
Metallic Bellows (India) — TamilCrew profile	https://tamilcrew.com/engineering/industrial/metal-bellows-expansion-joints/	**[U]**	AS9100 + ISO 9001; EJMA testing; assembly hardware range (gimbal, hinged, pressure-balanced)

Stage 9 — Clean, test & certify ★ the real semiconductor barrier

Source	URL	Grade	What it supports
KSM — cleaning & product cleanliness	https://www.ksmusa.com/cleaning-product-cleanliness	**[P]**	ISO Class 6 (1000) welding/assembly cleanroom; ISO Class 5 (100) final inspection/pack
KSM — semiconductor / vacuum R&D pages	https://www.ksmusa.com/semiconductor · https://www.ksmusa.com/vacuum-r-d	**[P]**	Capacity claims; rising vacuum valve demand context
Technetics — BELFAB edge welded bellows	https://technetics.com/products/belfab-edge-welded-metal-bellows/	**[P]**	Class 100/1000 cleanroom assembly
Witzenmann — edge welded bellows for high purity applications	https://media.witzenmann.com/mediapool/documents/brochures/edge-welded-bellows-for-high-purity-applications.pdf	**[P]**	UHV/UHP cleaning standard; optional RGA certificate
VAT Group — 04.2 high vacuum transfer valve	https://www.vatgroup.com/series/high-vacuum-transfer-valve-with-l-vat	**[P]**	Bellows material AISI 633 (AM350); leak rate $<1\times10^{-9}$ mbar·l/s spec — the qualification bar an Indian supplier would need to clear
Advanced Process Technology (APT)	https://aptglobal.com/vacuum-chambers/	**[P]**	The closest Indian analogue found — leak tests to 1×10^{-12} Torr·l/s, but for chambers, not bellows specifically
Machine Maker — KASFAB Tools facility	https://themachinemaker.com/news/kasfab-tools-opens-indias-first-semiconductor-equipment-manufacturing-facility-for-global-clients/	**[C]**	Confirms KASFAB has cleanrooms + precision welding, though not bellows-specific certification

Stage 10 — Component integration (valves, seals, feedthroughs)

Source	URL	Grade	What it supports
SMC — XGT high vacuum slit valve	https://www.smcusa.com/products/xgt-high-vacuum-slit-valve~166339	**[P]**	3 million cycle life; user-replaceable bellows; wafer window sizes
SMC — XGTP parallel seal slit valve	https://www.smcusa.com/products/xgtp-parallel-seal-slit-valve-for-high-vacuum~177109	**[P]**	Valve architecture and bellows-replacement access
SMC XGT slit valves datasheet	https://cdn.stevenengineering.com/wordpressprd/wp-content/uploads/2019/02/70_XGT-SLIT-VALVES-1.pdf	**[P]**	"Customer can replace bellows" — the consumable-part economics
VAT Group — 04.2 transfer valve	https://www.vatgroup.com/series/high-vacuum-transfer-valve-with-l-vat	**[P]**	Full valve spec, bellows material and leak-rate performance
Well Tech — mechanical seal bellows listings	(see Stage 7B)	**[U]**	Confirms Well Tech's output feeds directly into Indian mechanical seal makers at this stage

Stage 11 — OEM / equipment build

Source	URL	Grade	What it supports
Machine Maker — KASFAB Tools	https://themachinemaker.com/news/kasfab-tools-opens-indias-first-semiconductor-equipment-manufacturing-facility-for-global-clients/	**[C]**	India's first semiconductor equipment contract manufacturer; targets Applied Materials, Lam Research, Tokyo Electron, YES
Applied Materials India upstream equipment report	https://matribhumisamachar.com/en/2026/07/20/beyond-design-assembly-how-applied-materials-india-is-driving-the-nations-upstream-semiconductor-manufacturing-equipment-leap/	**[C]**	AMAT Bengaluru R&D scale; India Validation Center; supplier-development language
SEMI — Mid-Year Total Semiconductor Equipment Forecast	https://www.semi.org/en/semi-press-release/global-semiconductor-equipment-sales-forecast-to-reach-a-record-229-billion-dollars-in-2028-semi-reports	**[P]**	The global OEM demand base (Applied Materials, Lam, TEL, ASML, KLA et al.) this stage feeds

Stage 12 — End user & aftermarket (fabs, refineries, power, space)

Source	URL	Grade	What it supports
Indian Express — PM inaugurates Micron Sanand	https://indianexpress.com/article/cities/ahmedabad/pm-to-inaugurate-indias-first-semiconductor-fabrication-plant-worth-over-rs-20000-crore-in-sanand-10554477/	**[C]**	Micron ATMP operational; ₹22,516 cr; DRAM/NAND/SSD
YourStory — India semiconductor mission status	https://yourstory.com/2026/07/india-semiconductor-mission-37-years-interrupted	**[C]**	Tata-PSMC Dholera fab status; TSAT Assam; project table
Economic Times — Assam TSAT plant	https://economictimes.indiatimes.com/industry/cons-products/electronics/semiconductor-plant-in-assam-likely-to-start-production-this-fiscal-union-minister-ashwini-vaishnaw-says/articleshow/131421750.cms	**[C]**	TSAT Jagiroad capacity and investment
New Indian Express — Semicon India 2.0	https://www.newindianexpress.com/business/2026/Jul/15/cabinet-approves-rs-127-lakh-crore-for-semicon-india-20-to-boost-domestic-chip-ecosystem	**[C]**	Full project list: 12 approved units, ₹1.64 lakh cr
SEAJ — Worldwide SEMS Report (regional billings)	https://www.seaj.or.jp/english/statistics/4777967556197.pdf	**[P]**	Where semiconductor equipment is actually consumed globally (China/Taiwan/Korea 79% share) — the reason India is not yet a stage-12 anchor for bellows

Refinery, power, steel and space end-users (IOCL, RIL, BPCL, NTPC, NPCIL, SAIL, ISRO, BARC, DRDO) are sourced in `12-india-drivers-challenges.md` and `16-aerospace-defence-driver.md` — see those files for the government/ministry sources behind each sector's demand figures.

Cross-cutting: sourcing method and India import verification

These support the "who buys from whom" analysis threaded through stages 4, 6 and 7, rather than a single stage.

Source	URL	Grade	What it supports
Seair Exim Solutions — HS Code 7220 import data	https://www.seair.co.in/import-data-hs-code-7220.aspx	**[U]**	Sample customs entries: 0.08 mm Japanese SS foil at US$7.96–8.82/kg
Seair — HS Code 7219 (≥600mm width) import data	https://www.seair.co.in/cold-rolled-stainless-steel-coil-import-data/hs-code-7219.aspx	**[U]**	Commodity CR coil comparator pricing (US$1.87–2.39/kg)
Voleba — cold rolled stainless steel importer data	https://india-importers.voleba.com/products/india/cold-rolled-stainless-steel-importer-data.html	**[U]**	Gillette India / Vidyut Metallics importing 0.1 mm strip despite domestic razor-blade capacity — the closest analogue to the bellows-strip barrier

---

Iwatani's own capability (referenced across stages 4–6)

Source	URL	Grade	What it supports
Iwatani — Stainless Steel (Materials/ Metals)	https://www.iwatani.co.jp/eng/business/material/metals/products/stainless/	**[P]**	Ultrathin foil to 8 µm; precision slitting in Suzhou/ Zhongshan; spring, gasket, hydrogen-resistant, non-magnetic, duplex product lines
Iwatani — Metals (Materials)	https://www.iwatani.co.jp/eng/business/material/metals/	**[P]**	Full product-line list; Metals Department contact points
Iwatani — Metals network	https://www.iwatani.co.jp/eng/business/material/metals/network/	**[P]**	Domestic processing subsidiaries (Taihei Kozai, Kyushu Stainless Kako Center, Sanei Stainless Steel Center, Nose Kozai)
Iwatani — Corporate Brochure 2024	https://www.iwatani.co.jp/eng/common_v3/pdf/company_profile_2024.pdf	**[P]**	Domestic distribution network; welding demonstration room
Iwatani — FY2026 Investors' Guide	https://www.iwatani.co.jp/eng/ir/pdf/about_iwatani.pdf	**[P]**	Segment financials; Thailand/China integrated production precedent; 2024/2025 processing acquisitions

---

Summary: source count by stage

Stage	Distinct sources	Dominant grade
1. Raw materials	1	[P]
2. Melt & cast	6	[P]/[C]
3. Hot roll	1	[P]
4. Cold roll to thin gauge	**21**	mixed — the most heavily sourced stage
5. Temper & finish	0 (shares Stage 4/6 sources)	—
6. Precision slitting	8	[P]
7A. Formed bellows	11	[P]
7B. Edge-welded bellows	18	mixed [P]/[U]
8. Assembly & hardware	4	[P]
9. Clean, test & certify	7	[P]
10. Component integration	4	[P]
11. OEM / equipment build	3	[P]/[C]
12. End user & aftermarket	5	[C]/[P]
Cross-cutting sourcing	3	[U]
Iwatani capability	5	[P]

Total distinct sources behind the value chain map: ~97 (many appear in more than one stage, e.g. the Jindal Stainless brochure covers stages 2–4, VAT's valve page covers stages 7B, 9 and 10).

Read across: stages 4 and 7B carry the most sourcing weight, which is exactly right — they are the two stages where this pack makes its most load-bearing claims (India's precision-strip capability and tolerance gap; the existence and specification of Indian edge-welded bellows makers). Stage 9 is thin by design: the near-absence of sources reflects the near-absence of the capability in India, which is itself the finding.

For the complete list of every source used across the entire study (including market sizing, India policy, and the defence/aerospace deep dive not directly tied to the value chain), see `06-source-register.md`.